Answers and talk notes

Generate Equivalent Expressions

What your student is learning

Tonight we do two opposite moves: EXPANDING removes parentheses, and FACTORING puts them back.

Answers

Ask your student to explain before you look.

1. **C.** 3x + 6. The 3 reaches both terms: 3 × x + 3 × 2 = 3x + 6.
2. **A.** 5 × 2 + 5 × 7. The outside 5 multiplies each term inside: 5(2 + 7) = 5 × 2 + 5 × 7.
3. **B.** 6. The largest number that divides both 12 and 18 is 6.
4. **D.** 8x + 6. Multiply each term by 2: 2 × 4x = 8x and 2 × 3 = 6, so 8x + 6.
5. **A.** 4(2x + 3). The GCF of 8 and 12 is 4, and 8x ÷ 4 = 2x and 12 ÷ 4 = 3, so 4(2x + 3).
6. **C.** 3x + 12. The two parts have areas 3 × x = 3x and 3 × 4 = 12, so the total is 3x + 12.

Talk about it

Did the outside number reach every term inside the parentheses? What is the greatest common factor (GCF) of the two terms?

Home mission

Set out 4 plates. Put 3 crackers and 2 grapes on each. Can your student say the total two ways, 4(3 + 2) and 4 × 3 + 4 × 2? Then push the food together: 12 crackers and 8 grapes. What number is shared by both?

When they are stuck, ask

- Did the outside number reach every term inside?
- What is the biggest number that divides both terms?
- Can you check by doing the opposite move?

Try not to say

- Multiply the outside number by the first term only.
- Any shared factor works, it does not have to be the greatest.
- Expanding and factoring are unrelated skills.